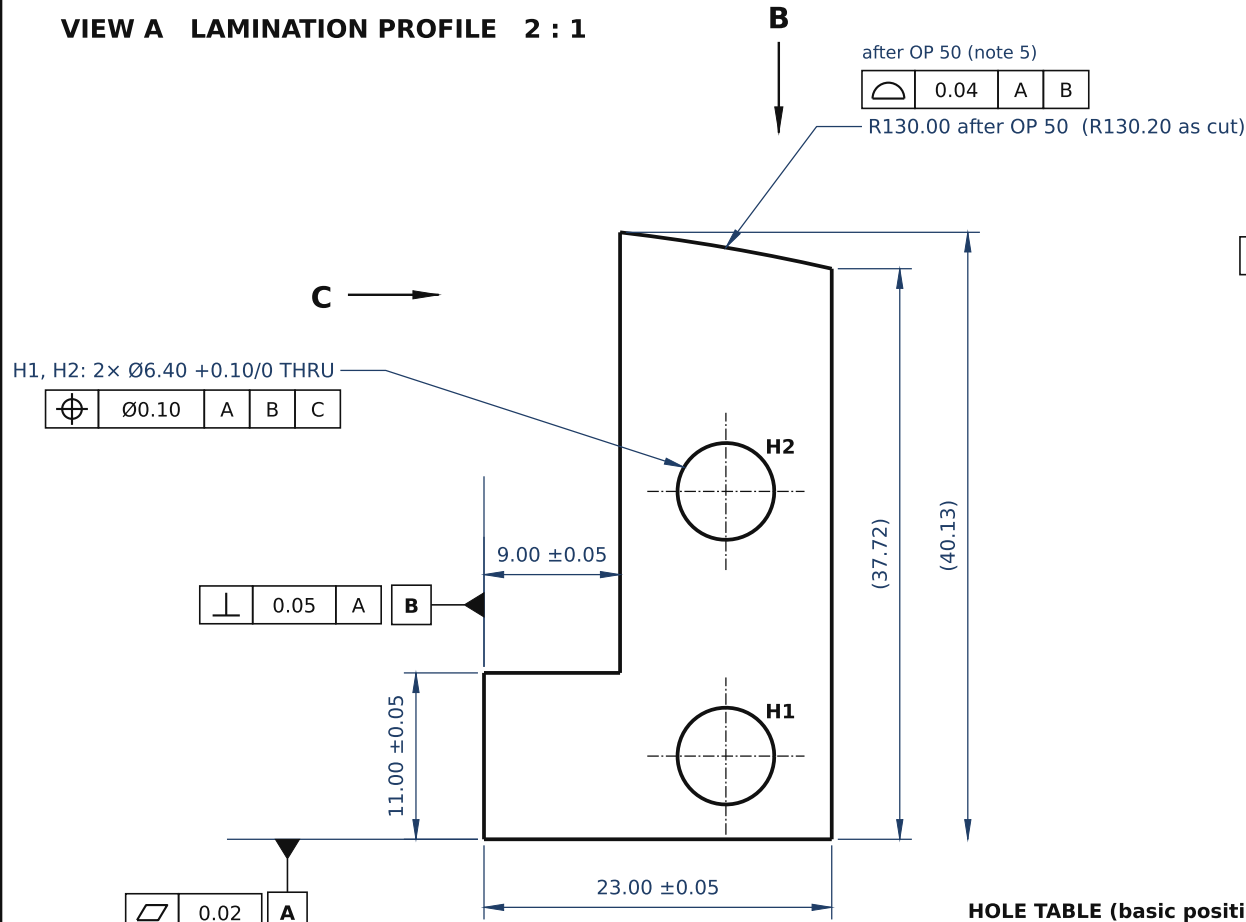


VIEW A LAMINATION PROFILE 2 : 1



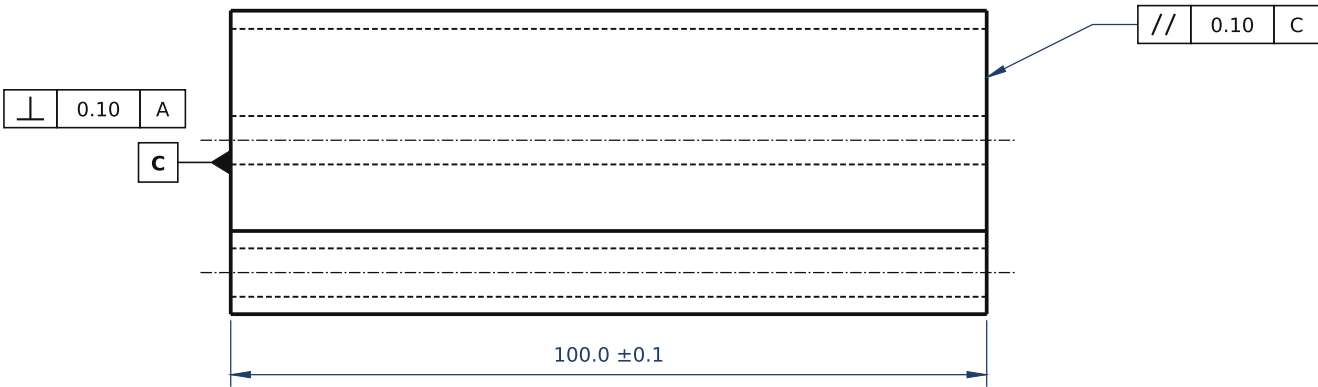
HOLE TABLE (basic positions)

HOLE	X from B	Y from A	SIZE
H1	16.00	5.50	Ø6.40 +0.10/0 THRU
H2	16.00	23.00	Ø6.40 +0.10/0 THRU

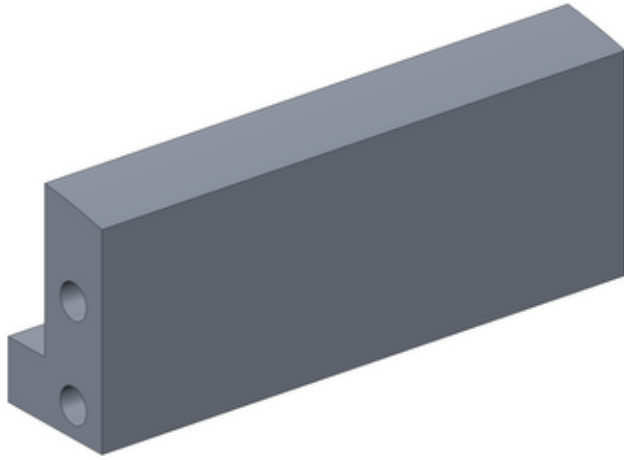
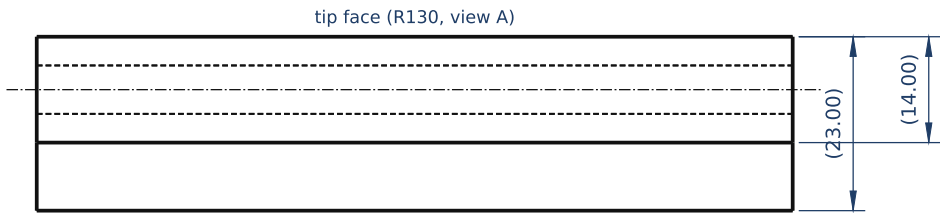
NOTES

- 1 Material: non-oriented electrical steel M235-35A to EN 10106, 0.35 mm, fully processed. Insulation coating on both sides: C-5 (EN 10342) for an epoxy-bonded stack, or the supplier's self-bonding varnish for a varnish-bonded stack.
- 2 Laminations: laser or wire-EDM cut to the DXF DCCREG-UTR-101_lamination.dxf (as-cut profile, tip face R130.20). Burr ≤ 0.02 mm, all burrs to one side within a stack. No interlocks, rivets or welds: the two $\varnothing 6.40$ holes are the only through-features.
- 3 Stack: 100.0 ± 0.1 mm at a stacking factor ≥ 0.95 ; laminations aligned on datums A and B and the two holes; bonded (self-bonding varnish cured to the supplier's cycle, or vacuum epoxy). Hand B is stacked from laminations turned over.
- 4 Stress-relief anneal after cutting (optional, to agree): 750°C for 2 h in a non-oxidising atmosphere, before bonding. Epoxy route only: a self-bonding varnish does not survive the anneal.
- 5 Tip face: supplied as cut to R130.20 (0.20 mm stock). OP 50 (by DCCREG, on the assembled rotor) finishes it to R130.00: wire EDM or a light grind, then deburr and re-varnish the face. The profile tolerance on the tip face applies after OP 50. The centre of R130 is the machine axis: basic 6.00 from B (away from the tip) and 89.00 below A.
- 6 Face A seats the NiFe neck strip (a lap joint in the magnetic circuit): keep A flat, clean and free of burrs.
- 7 Marking: hand (A / B) and stack number on the tip's outer face ($x' = 23$) by paint or laser; never on A, B, C or the tip face.
- 8 Inspection: profile and holes on the bonded stack (CMM or optical); stack height; interlaminar insulation by sample (Franklin test, IEC 60404-11).
- 9 General tolerances ISO 2768-mK. Dimensions in mm. Break sharp edges 0.1 max, except the tip-face edges (keep sharp).
- 10 Views by reference arrows (ISO 128-30). Boxed values are basic dimensions. Do not scale from this drawing.

VIEW C 1 : 1



VIEW B 1 : 1



3-D VIEW, hand A: the L-profile end face, the outer face and the tip face (axonometric, not to scale)

LAMINATION / STACK DATA	
lamination	one part for both hands (hand B = laminations turned over)
thickness, material	0.35 mm, M235-35A (EN 10106), coated both sides
stack	100.0 mm, $\approx$ 271 laminations at a stacking factor $\geq$ 0.95
section, mass	5.81 cm ² net, 0.42 kg per stack
per machine	6 stacks hand A + 6 stacks hand B (3 utrons per side, 2 sides)
mates with	NiFe neck strip on A, G10 spacer on B, G10 cheeks on C and its opposite end

REVISIONS		
Rev	Date	Description
A	2026-10-08	First issue, draft for quotation

DCCREG turbine · wound utron, magnetic doubler (rotor)		
HALF-CORE, LAMINATED (hand A; hand B mirror)		Drawing no.
Material: M235-35A, 0.35 mm (EN 10106), bonded stack		DCCREG-UTR-101
Mass: 0.42 kg	Units: mm	Rev A (draft) Sheet 1 / 1 A3
Scale: 2:1 (A), 1:1 (B, C)	Tolerances: ISO 2768-mK	Date: 2026-10-08
Drawn: generated from sim/utron_profile.py	Checked: — Approved: —	DRAFT FOR QUOTATION